

Independent Validation of Transgenerational Inheritance of Learned Pathogen Avoidance in *Caenorhabditis elegans*

Reviewed Preprint

v1 • June 13, 2025

Not revised

Aalimah Akinosho, Joseph Alexander, Kyle Floyd, Andrés Vidal-Gadea 

School of Biological Sciences, Illinois State University, Normal, United States

 https://en.wikipedia.org/wiki/Open_access Copyright information

eLife Assessment

This **valuable** study concerns a model for transgenerational epigenetic inheritance, the learned avoidance by *C. elegans* of the PA14 pathogenic strain of *Pseudomonas aeruginosa*. A recent study questioned whether transgenerational inheritance in this paradigm lacks robustness. The authors of this study have worked independently of the group that reported the original phenomenon and also independently of the group that challenged the original report. With **solid** data, this study independently validates findings previously reported by the Murphy group, confirming that the paradigm is reproducible elsewhere. The present study is therefore of broad interest to anyone studying genetics, epigenetics, or learned behavior.

<https://doi.org/10.7554/eLife.107034.1.sa3>

Abstract

Transgenerational inheritance of learned behaviors remains a controversial topic in biology. The Murphy lab previously demonstrated that *C. elegans* exposed to pathogenic *Pseudomonas aeruginosa* (PA14) not only learn to avoid this bacterium but also transmit this avoidance behavior to untrained offspring. A recent study has challenged these findings, questioning the robustness of this phenomenon. Here, we independently validate the transgenerational inheritance of PA14 avoidance in *C. elegans*. Using standardized choice assays, we show that worms trained on PA14 develop significant avoidance that persists in F1 and F2 generations, though with decreased strength after P0. Our results provide independent confirmation of transgenerational inheritance of PA14 avoidance through the F2 generation, in agreement with previous findings from the Murphy lab and in contrast to a recent study by the Hunter group that failed to detect this phenomenon beyond the F1 generation.

Introduction

Caenorhabditis elegans inhabits soil environments where it encounters diverse bacteria, including both nutritional and pathogenic species (Schulenburg and Félix, 2017 [↗](#); Neher, 2010 [↗](#)). This nematode has evolved sophisticated sensory mechanisms to distinguish beneficial from harmful bacteria (Kim and Flavell, 2020 [↗](#); Guillermin, 2018 [↗](#)). Interestingly, naïve *C. elegans* initially prefer the pathogenic bacterium *Pseudomonas aeruginosa* (PA14) over the non-pathogenic laboratory strain *Escherichia coli* (OP50) but rapidly learn to avoid PA14 following exposure (Zhang and Bargmann, 2005).

The Murphy lab first demonstrated that this learned avoidance behavior can be transmitted to progeny, persisting through the F2 generation (Moore et al. Cell 2019; Kaletsky et al. Nature 2020; Moore et al. Cell 2021; Sengupta et al., 2024 [↗](#)). This transgenerational epigenetic inheritance (TEI) has been linked to small RNA pathways and the dsRNA transport proteins SID-1 and SID-2. Recently, Sengupta et al. (2024) [↗](#) further identified a specific small RNA responsible for inducing transgenerational inheritance of learned avoidance.

However, Gainey et al. (2025) [↗](#), representing the Hunter group, reported that while parental and F1 avoidance behaviors were evident, transgenerational inheritance was not reliably observed beyond the F1 generation under their experimental conditions. These conflicting findings highlight the importance of methodological details and underscore the need for independent replication. Here, we rigorously examine whether parental exposure to PA14 elicits consistent avoidance in subsequent generations, addressing potential methodological variations that might influence results. Our study specifically focuses on the transmission of learned avoidance through the F2 generation—beyond the intergenerational (F1) effect—because this is where divergence between published studies begins. Our goal was to evaluate whether the reported transgenerational signal persists across two generations without reinforcement.

Results and Discussion

We investigated whether *C. elegans* exposed to PA14 exhibit learned avoidance behavior that persists across generations. Naïve worms showed a preference for PA14 over OP50, with a negative choice index (approximately -0.4). After 24 hours of exposure to PA14, trained worms developed strong aversion to the pathogen, exhibiting a significantly positive choice index (**Figure 1** [↗](#)).

Critically, this avoidance behavior persisted in the untrained F1 and F2 progeny of trained worms, with both generations showing significantly higher choice indices compared to naïve controls (**Fig. 1** [↗](#), Table S1). However, the strength of avoidance diminished somewhat in F2 relative to F1 progeny, suggesting a gradual dilution of the inherited response over generations.

Our findings independently validate the transgenerational inheritance of learned pathogen avoidance initially reported by the Murphy lab. In contrast, Gainey et al. (2025) [↗](#) failed to observe such inheritance beyond F1 despite exhaustive attempts and protocol refinements. While we cannot definitively reconcile these differences, our results suggest that under tightly controlled conditions—including bacterial lawn density, OD₆₀₀=1.0 standardization, and immobilization with sodium azide restricted to one hour (to capture initial preference behaviors)—transgenerational inheritance through F2 is both detectable and statistically robust.

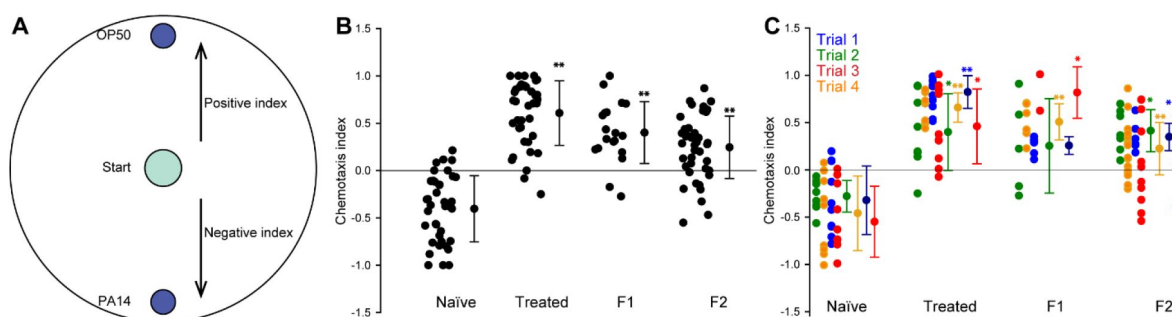


Figure 1.

Transgenerational inheritance of PA14 avoidance in *C. elegans*.

A. Chemotaxis assay setup. A 25 μ L drop of OP50 and PA14 bacteria of the same optical density ($OD_{600} = 1$) plus 1.0 μ L of 400 mM NaN₃ were placed equidistant from the center of a 10cm chemotaxis plate where worms were placed at the start of the assay (see methods). indices were measured for naïve worms, trained worms, first-generation (F1), and second-generation (F2) offspring. **B.** Pooled results from four independent trials showing that Trained worms, which were exposed to PA14 for 24 hours, exhibited significantly higher choice indices, indicating learned avoidance of PA14. This aversion persisted in F1 and F2 progeny, demonstrating transgenerational inheritance of pathogen avoidance behavior. **C.** Comparison of data separated into individual trials. Ranked summed test, means and SEM shown (see Table S1 for statistics).

We echo Gainey et al.'s observation that behavioral assays can be highly sensitive to environmental variables. This highlights the importance of consistency across experimental setups and support the view that context-dependent variation may underlie previously reported discrepancies.

These results support a growing body of evidence that *C. elegans* can transmit acquired behavioral adaptations to offspring through non-genetic mechanisms, likely involving small RNAs and epigenetic modifications as suggested by Sengupta et al. (2024) [↗](#). The gradual decrease in avoidance strength by the F2 generation suggests that the inherited signal attenuates over successive generations without reinforcement.

Methods

Nematode Strains and Maintenance

Wild-type *Caenorhabditis elegans* (N2) were maintained on nematode growth medium (NGM) plates seeded with *Escherichia coli* OP50 at 20°C under standard conditions. Worm populations were passaged regularly to avoid overcrowding and starvation.

Bacterial Cultures

We inoculated 6 mL of LB medium (pH 7.5) with single colonies of OP50 or *Pseudomonas aeruginosa* PA14 and incubated them overnight at 37°C with shaking at 250 rpm. For PA14, cultures did not exceed 16 hours of incubation or develop visible biofilms. For both OP50 and PA14, the overnight culture was diluted to an OD₆₀₀ of 1.0 in fresh LB media prior to seeding.

Synchronization of Worms

Gravid adults were collected from OP50-seeded NGM plates by washing with NGM buffer. After settling by gravity, a standard bleach solution was added and gently nutated for 5–10 minutes. The embryo pellet was washed 2–3 times with NGM after centrifugation. Approximately 250–350 eggs were plated onto each OP50-seeded NGM plate and incubated at 20°C for 48–52 hours until reaching late L4 stage.

Training and Choice Assay

Training plates (10 cm NGM) were seeded with 1 mL of overnight bacterial cultures and incubated at 25°C for 2 days. Choice assay plates (6 cm NGM) were prepared with 25 µL spots of OP50 and PA14 on opposite sides placed 24 hrs prior to the assays. Late L4 worms were transferred to training plates (OP50 or PA14) for 24 hours at 20°C. For choice assays, 1.0 µL of 400 mM sodium-azide was placed on each bacterial spot before adding worms. After one hour, the number of worms immobilized on each spot was counted to calculate a Choice Index: Choice Index = (fraction of worms on OP50) - (fraction of worms on PA14).

Transgenerational Testing

Synchronized F1 progeny were obtained by bleaching trained adult worms and allowing embryos to develop on OP50-seeded plates at 20°C. On Day 1 of adulthood, F1 worms were tested using the same choice assay. The F2 generation was derived from untrained F1 adults, synchronized in the same manner, and tested in identical conditions.

Controls and Additional Considerations

The data presented in this study are the result of four independent rounds of experimentations. Each individual assay was performed with animals harvested from unique culture plates. An average of 55 animals participated in each assay (were immobilized in the NaN3 and tallied at the end of the assay). We conducted an average of 8.5 assays per condition during each of the four replicates. All plates used in the study were prepared at least two days before experiments to ensure consistent bacterial lawn growth. Worms were monitored to prevent starvation or overcrowding, as these conditions could influence their bacterial preferences and behavioral responses. Fresh bacterial stocks were maintained at -80°C and streaked weekly to ensure culture consistency.

Statistical Analysis

Graphs and statistical comparisons were conducted using Sigmaplot 14 (Inpixon). We used the InterQuartile Range ($>1.5 \times \text{IQR}$) method to evaluate all data. Exclusion of outliers (5/143 data points: 3.5%) did not affect statistical significance or effect. Comparisons involved One Way ANOVAS when dataset passed normality (Shapiro-Wilk), and equal variance (Brown-Forsythe) tests. Failed either of these comparisons involved non-parametric tests (Kruskal-Wallis One Way Analysis of Variance on Ranks). Post hoc comparisons involved Multiple Comparisons versus Control Group (Holm-Sidak method) following ONE Way ANOVAS, or Multiple Comparisons versus Control Group (Dunn's Method) following Kruskal-Wallis One Way Analysis of Variance on Ranks (See Table S1 for statistical data, and Supp. File 1 for measurements).

Supplementary Information

TRIAL	Naïve	Treated	F1	F2	H	DF	P	Dunn's Naïve - Treated			Dunn's Naïve - F1			Dunn's Naïve - F2		
	N	N	N	N				Diff Rank	Q	P	Diff Rank	Q	P	Diff Rank	Q	P
1	10	12	6	7	29.23	3	<0.001	23.5	5.4	<0.001	9.7	1.8	0.2	12.3	2.4	0.044
2	8	8	5	7	12.8	3	0.005	12.4	3	0.008	10.1	2.2	0.09	13.1	3.1	0.006
3	10	10	2	11	18	3	<0.001	16.6	3.8	<0.001	21.4	2.9	0.013	10	2.4	0.053
					Shapiro-Wilk	Brown-Forsythe	Alpha-Power	Holm-Sidak Naïve - Treated			Holm-Sidak Naïve - F1			Holm-Sidak Naïve - F2		
								Diff Mean	t	P	Diff Mean	t	P	Diff Mean	t	P
4	10	8	6	18	0.48	0.084	0.05/1.00	1.1	8.3	<0.001	0.97	6.6	<0.001	0.68	6.1	<0.001
	Naïve	Treated	F1	F2	H	DF	P	Dunn's Naïve - Treated			Dunn's Naïve - F1			Dunn's Naïve - F2		
								Diff Rank	Q	P	Diff Rank	Q	P	Diff Rank	Q	P
Pooled	38	38	19	43	77.6	3	<0.001	78.4	8.5	<0.001	60.6	5.4	<0.001	46.9	5.3	<0.001

Table S1.

Summary of statistical analysis.

H: H-statistic; DF: degrees of freedom; Q: standardized difference between ran sums; Trial: Independent trials conducted on different dates; N: independent assays conducted on a trial (each trial derived from individual culture plate); **Naïve**: Day-1 adults without previous exposure to PA14; **Treated**: Day-1 adults with previous exposure to PA14; **F1**: Day-1 adults without previous exposure to PA14 derived from **Treated** group; **F2**: Day-1 adults without previous exposure to PA14 derived from **F1** group.

Acknowledgements

We thank Dr. Murphy at Princeton University for reagents and protocols. Some strains were provided by the CGC, which is funded by NIH Office of Research Infrastructure Programs (P40 OD010440).

Additional information

Funding

Funding was provided by the National Institutes of Health, National Institute of Arthritis and Musculoskeletal and Skin Diseases award 2R15AR068583-02.

References

- Baugh L. R., Day T. (2020) **Nongenetic inheritance and multigenerational plasticity in the nematode *C. elegans*** *eLife* **9**:e58498 <https://doi.org/10.7554/eLife.58498> | [Google Scholar](#)
- Cecere G. (2021) **Small RNAs in epigenetic inheritance: from mechanisms to trait transmission** *FEBS letters* **595**:2953–2977 [Google Scholar](#)
- Diaz S. A., Mooring E. Q., Rens E. G., Restif O. (2015) **Association with pathogenic bacteria affects life-history traits and population growth in *Caenorhabditis elegans*** *Ecology and evolution* **5**:1653–1663 [Google Scholar](#)
- Gainey D. P., Shubin A. V., Hunter C. P. (2025) **Reported transgenerational responses to *Pseudomonas aeruginosa* in *Caenorhabditis elegans* are not robust** *eLife* **13**:RP100254 <https://doi.org/10.7554/eLife.100254> | [Google Scholar](#)
- Guillermin M. (2018) **Exploring the neural basis of chemosensory behaviors in *Caenorhabditis elegans*: How context and experience shape sensory perception** UCLA [Google Scholar](#)
- Kaletsky R., Moore R. S., Vrla G. D., Parsons L. R., Gitai Z., Murphy C. T. (2020) ***C. elegans* interprets bacterial non-coding RNAs to learn pathogenic avoidance** *Nature* **586**:445–451 [Google Scholar](#)
- Kim D. H., Flavell S. W. (2020) **Host-microbe interactions and the behavior of *Caenorhabditis elegans*** *Journal of Neurogenetics* **34**:500–509 [Google Scholar](#)
- Moore R. S., Kaletsky R., Murphy C. T. (2019) **Piwi/PRG-1 argonaute and TGF- β mediate transgenerational learned pathogenic avoidance** *Cell* **177**:1827–1841 [Google Scholar](#)
- Moore R. S. (2021) ***C. elegans* Intergenerational and Transgenerational Responses to *P. Aeruginosa*** Princeton University [Google Scholar](#)
- Neher D. A. (2010) **Ecology of plant and free-living nematodes in natural and agricultural soil** *Annual review of phytopathology* **48**:371–394 [Google Scholar](#)
- Schulenburg H., Félix M. A. (2017) **The natural biotic environment of *Caenorhabditis elegans*** *Genetics* **206**:55–86 [Google Scholar](#)
- Sengupta T., St. Ange J., Kaletsky R., Moore R. S., Seto R. J., Marogi J., ..., Murphy C. T. (2024) **A natural bacterial pathogen of *C. elegans* uses a small RNA to induce transgenerational inheritance of learned avoidance** *PLoS genetics* **20**:e1011178 [Google Scholar](#)
- Swierczek N. A., Giles A. C., Rankin C. H., Kerr R. A. (2011) **High-throughput behavioral analysis in *C. elegans*** *Nature methods* **8**:592–598 [Google Scholar](#)
- Zhang Y., Lu H., Bargmann C. I. (2005) **Pathogenic bacteria induce aversive olfactory learning in *Caenorhabditis elegans*** *Nature* **438**:179–184 [Google Scholar](#)

Author information

Aalimah Akinosho*

School of Biological Sciences, Illinois State University, Normal, United States

* contributed equally to this work

Joseph Alexander*

School of Biological Sciences, Illinois State University, Normal, United States

* contributed equally to this work

Kyle Floyd

School of Biological Sciences, Illinois State University, Normal, United States

Andrés Vidal-Gadea

School of Biological Sciences, Illinois State University, Normal, United States

ORCID iD: [0000-0001-5981-5528](https://orcid.org/0000-0001-5981-5528)

For correspondence: avidal@ilstu.edu

Editors

Reviewing Editor

Detlef Weigel

Max Planck Institute for Biology Tübingen, Tübingen, Germany

Senior Editor

Detlef Weigel

Max Planck Institute for Biology Tübingen, Tübingen, Germany

Reviewer #1 (Public review):

Summary:

The manuscript addresses the discordant reports of the Murphy (Moore et al., 2019; Kaletsky et al., 2020; Sengupta et al., 2024) and Hunter (Gainey et al., 2025) groups on the existence (or robustness) of transgenerational epigenetic inheritance (TEI) controlling learned avoidance of *C. elegans* to *Pseudomonas aeruginosa*. Several papers from Colleen Murphy's group describe and characterize *C. elegans* transgenerational inheritance of avoidance behaviour. In the hands of the Murphy group, the learned avoidance is maintained for up to four generations, however, Gainey et al. (2025) reported an inability to observe inheritance of learned avoidance beyond the F1 generation. Of note, Gainey et al used a modified assay to measure avoidance, rather than the standard assay used by the Murphy lab. A response from the Murphy group suggested that procedural differences explained the inability of Gainey et al.(2025) to observe TEI. They found two sources of variability that could explain the discrepancy between studies: the modified avoidance assay and bacterial growth conditions (Kaletsky et al., 2025). The standard avoidance assay uses azide as a paralytic to capture worms in their initial decision, while the assay used by the Hunter group does not capture the

worm's initial decision but rather uses cold to capture the location of the population at one point in time.

In this short report, Akinosho, Alexander, and colleagues provide independent validation of transgenerational epigenetic inheritance (TEI) of learned avoidance to *P. aeruginosa* as described by the Murphy group by demonstrating learned avoidance in the F2 generation. These experiments used the protocol described by the Murphy group, demonstrating reproducibility and robustness.

Strengths:

Despite the extensive analyses carried out by the Murphy lab, doubt may remain for those who have not read the publications or for those who are unfamiliar with the data, which is why this report from the Vidal-Gadea group is so important. The observation that learned avoidance was maintained in the F2 generation provides independent confirmation of transgenerational inheritance that is consistent with reports from the Murphy group. It is of note that Akinosho, Alexander et al. used the standard avoidance assay that incorporates azide, and followed the protocol described by the Murphy lab, demonstrating that the data from the Moore and Kaletsky publications are reproducible, in contrast to what has been asserted by the Hunter group.

Weaknesses:

While I would have liked to see a confirmation of the *daf-7::GFP* data in F2, and perhaps inheritance of avoidance beyond F2, the premise of the manuscript is that they have independently verified the transgenerational inheritance of learned avoidance as described by the Murphy lab, and this bar has been met.

<https://doi.org/10.7554/eLife.107034.1.sa2>

Reviewer #2 (Public review):

Summary:

The manuscript "Independent validation of transgenerational inheritance of learned pathogen avoidance in *C. elegans*" by Akinosho and Vidal-Gadea offers evidence that learned avoidance of the pathogen PA14 can be inherited for at least two generations. In spite of initial preference for the pathogen when exposed in a 'training session', 24 hours of feeding on this pathogen evoked avoidance. The data are robust, replicated in 4 trials, and the authors note that diminished avoidance is inherited in generations F1 and F2.

Strengths:

These results contrast with those reported by Gainey et al, who only observed intergenerational inheritance for a single generation. Although the authors' study does not explain why Gainey et al fail to reproduce the Murphy lab results, one possibility is that a difference in a media ingredient could be responsible.

Weaknesses:

The authors do not list the sources of their media ingredients, which might be important with regard to reproducibility.

<https://doi.org/10.7554/eLife.107034.1.sa1>

Reviewer #3 (Public review):

Summary

This short paper aims to provide an independent validation of the transgenerational inheritance of learned behaviour (avoidance) that has been published by the Murphy lab. The robustness of the phenotype has been questioned by the Hunter lab. In this paper, the authors present one figure showing that transgenerational inheritance can be replicated in their hands. Overall, it helps to shed some light on a controversial topic.

Strengths

The authors clearly outline their methods, particularly regarding the choice of assay, so that attempting to reproduce the results should be straightforward. It is nice to see these results repeated in an independent laboratory.

Weaknesses

Previous reports on this topic have provided raw data, which is helpful when assessing sample sizes. The authors provided a spreadsheet containing the choice assay results for individual assays, but not the raw data. In the methods, it is stated that F2 animals were produced from F1 animals by bleaching, but there are many more F2 assays than F1. Were multiple F2 assays performed on the offspring from one F1 plate? If so, they do not represent independent assays.

I think that the introduction somewhat overstates their findings - do they really "address potential methodological variations that might influence results"? This makes it sound as though they test different conditions, whereas they only use one assay setup throughout.

<https://doi.org/10.7554/eLife.107034.1.sa0>